

## Exercise 03: Asymptotic Notation

Write a text file named '`notation.txt`' that proves each equation below

### Equations

1.  $5n - 2 = \Theta(n)$
2.  $3n^2 + 6n + 2 = \Theta(n^2)$
3.  $10n + 100 = O(n^2)$
4.  $9n - 4 = \omega(1)$
5.  $n^3 - 6n = \Theta(n^3)$
6. If  $f(n) = O(g(n))$  and  $g(n) = O(h(n))$ , then  $f(n) = O(h(n))$ .